

CONTROLLED UNCLASSIFIED INFORMATION (CUI)

Secure File Encryption for Email Transmission

Before You Begin: Choose the Right Method

OPTION 1: S/MIME Only (Both Parties Have PIV/CAC)

If you have your recipient's S/MIME certificate, you can use S/MIME alone:

1. Compose your email in Outlook with CUI files attached
2. Click **Sign** and **Encrypt** before sending

OPTION 2: S/MIME + SendCUIEmail (Maximum Protection)

For defense-in-depth, use **both** S/MIME and SendCUIEmail together:

- Files are encrypted with a password (content protection)
- Email is signed and encrypted with S/MIME (transport protection)
- Password sent separately (out-of-band delivery)

Why? If email is accidentally forwarded or misdirected, files remain encrypted. Separating password from content minimizes accidental leakage.

OPTION 3: SendCUIEmail Only – Use when recipient lacks PKI capability (no PIV/CAC/S/MIME), such as external partners or contractors.

TIP: Always **sign your email** if you have a PIV/CAC – this proves it came from you.

Part A: Encrypting Files (Sender)

Step 1: Launch the Tool

1. Navigate to the SendCUIEmail folder
2. Double-click **Encrypt.bat**
3. A blue PowerShell window will open

Step 2: Select Files

1. When prompted, drag files into the window OR type the file path
2. Press Enter when done adding files

Step 3: Choose CUI Category

1. Select the appropriate CUI marking (0-4)
2. Press Enter for basic CUI if unsure

Step 4: Create Password

1. Type a password (minimum 8 characters) OR press Enter for auto-generated passphrase

2. **IMPORTANT:** Characters will NOT appear as you type – this is normal!
3. If using auto-generated passphrase, copy it down immediately
4. Otherwise, type your password again to confirm
5. Remember this password/passphrase – you’ll share it separately

Step 5: Send the Files

1. Find the output folder (path shown in yellow)
2. Attach the .Locked files and README.md to your email
3. If you have a PIV/CAC, click **Sign** before sending (proves it’s from you)
4. Send the email to your recipient

WARNING: Password Delivery

COMPLIANT: Phone call, SMS text, in-person, or S/MIME encrypted email

NOT COMPLIANT: Unencrypted email (even if separate from files)

Part B: Decrypting Files (Recipient)

Step 1: Save the Files

1. Download all `.Locked` files from the email
2. Download the `README.md` file
3. Save them all to the same folder (e.g., Desktop)

Step 2: Open PowerShell

1. Press **Windows Key + X**
2. Click “Windows PowerShell” or “Terminal”
3. A blue command window will open

Step 3: Run the Decryption Command

1. Open the `README.md` file (double-click or use Notepad)
2. Find the “PowerShell One-Liner” section
3. Copy the entire command (starts with `$f=Read-Host`)
4. Paste it into PowerShell and press Enter

Step 4: Enter File Path and Password

1. When prompted for “File”, type the full path to a `.Locked` file
Example: C:\Users\YourName\Desktop\Document.pdf.Locked
2. When prompted for “Password”, type the password you received
3. **IMPORTANT:** Characters will NOT appear – this is normal!

Step 5: Verify Decryption

1. The decrypted file appears in the same folder
2. *Example: Document.pdf.Locked becomes Document.pdf*
3. Repeat steps 3-4 for each `.Locked` file

TIP: Use the same password for all files in a batch.
If decryption fails, double-check the password and file path.